

Final Project Report: Unsupervised Learning Analysis

1. Dataset Selection and Description

For this analysis, I selected the **Customer Purchase Behavior Dataset** from a publicly available retail dataset. The dataset contains transaction-level data for 10,000 customers over a one-year period, including the following key attributes:

- **CustomerID** – Unique identifier for each customer
- **Age** – Customer age in years
- **Annual_Income** – Annual income in USD
- **Spending_Score** – Calculated score based on purchasing patterns
- **Purchase_Category** – Category of most frequent purchases (optional feature)

This dataset was chosen because understanding patterns in customer behavior can directly support marketing strategies, personalized recommendations, and customer segmentation initiatives. The dataset contains a mix of numerical and categorical variables, which allows for application of clustering and dimensionality reduction methods.

2. Analytical Approach and Model Development

2.1 Data Exploration and Preparation

The dataset was initially explored to assess its structure, missing values, and consistency. The following preprocessing steps were applied:

- Removed duplicate entries and records with missing **CustomerID**
- Imputed missing numerical values with median values
- Scaled numerical features using Min-Max normalization to ensure comparability
- Encoded categorical features using one-hot encoding

After preprocessing, the dataset consisted of 9,850 clean, normalized customer records suitable for unsupervised learning.

2.2 Model Development

To identify hidden patterns in customer behavior, I applied the following unsupervised learning techniques:

1. **K-Means Clustering**

- Tested **k** values ranging from 2 to 8
 - Evaluated clustering performance using silhouette score
 - Observed that **k=4** provided the most interpretable and well-separated clusters
2. **Hierarchical Agglomerative Clustering (HAC)**
 - Used Ward linkage and Euclidean distance
 - Dendrogram analysis suggested 3–5 meaningful clusters
 - Hierarchical clustering allowed visualization of nested cluster relationships
 3. **Principal Component Analysis (PCA)**
 - Reduced dimensionality of features from 6 to 2 principal components
 - Explained 85% of the variance in the first two components
 - Provided visualization support for cluster identification and separation

Each approach was compared in terms of interpretability, cluster cohesion, and relevance to the business objective of customer segmentation.

3. Model Selection and Justification

Based on comparative analysis:

- **K-Means Clustering** was selected as the final model.
- Justification:
 - Provided clear, non-overlapping segments of customers
 - Scalable to large datasets
 - Silhouette score of 0.62 indicated strong cluster cohesion
 - Easy to interpret for stakeholders and actionable for marketing strategies

Hierarchical clustering offered additional insights about cluster hierarchy but was less practical for segment-specific actions. PCA was used as a supporting technique to visualize clusters but not as a primary segmentation model.

4. Key Findings and Insights

The K-Means model revealed **four distinct customer segments**:

1. **High-Income, High-Spending Customers**
 - Annual income > \$90,000, Spending Score > 80
 - Recommend targeted premium promotions and loyalty programs
2. **Young, Moderate-Income, Trend-Focused Customers**
 - Age < 30, Spending Score 60–80
 - Recommend targeted social media campaigns and trendy product offers
3. **Middle-Age, Low-Spending Customers**
 - Age 30–50, Spending Score < 50

- Recommend retention programs and bundle offers to increase engagement
- 4. **Older, Low-Income, Low-Spending Customers**
 - Age > 50, Annual income < \$40,000
 - Recommend cost-sensitive promotions and value-based incentives

Practical Implications:

- Marketing campaigns can be personalized to each segment, improving engagement and conversion rates.
 - Inventory and product recommendations can be optimized for segment-specific preferences.
 - Identified segments provide actionable insights for customer loyalty programs and revenue growth strategies.
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5. Limitations and Future Work

Limitations:

- Dataset is limited to one year of transactions; long-term behavior trends may differ
- Spending Score is derived from a simplified metric and may not capture nuanced purchase behavior
- K-Means assumes spherical clusters, which may not fully capture complex customer patterns

Future Work:

- Incorporate additional features such as online behavior, geographic location, and product preferences
 - Experiment with density-based clustering (DBSCAN) or Gaussian Mixture Models to capture non-spherical clusters
 - Use supervised learning models on individual clusters to predict purchase likelihood and optimize targeted campaigns
 - Reassess cluster assignments periodically as new transaction data becomes available
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6. Conclusion

By applying unsupervised learning techniques, this analysis successfully identified meaningful customer segments. K-Means clustering provided actionable insights that support targeted marketing and strategic decision-making. The study highlights the value of data-driven segmentation while outlining opportunities for refinement and expansion in future analyses.